OptiscanAI: Large Vision Models for Multi-Disease Retinal Screening

A Benchmark and Deployment Study in Ugandan Healthcare • Paper 83

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Three Core Realities in Retinal AI

01
The Clinical Tragedy
2.2 billion people suffer visual impairment globally. In Uganda, <45 specialists serve 48 million citizens. The doctors are in Kampala; the blindness is in rural districts.

02
The Benchmark Illusion
A 93% PPV on an enriched international benchmark collapses to 16% in rural primary care. We cannot confuse academic leaderboards with clinical readiness.

03
The Silent Defect
Severe probability over-confidence (ECE 0.209) silently disabled 5 whole disease channels. We confront this failure head-on and prove the isotonic remedy.

“Before deploying AI into human healthcare, we have a moral obligation to state where it breaks down — and engineer the remedy before it reaches the clinic.”

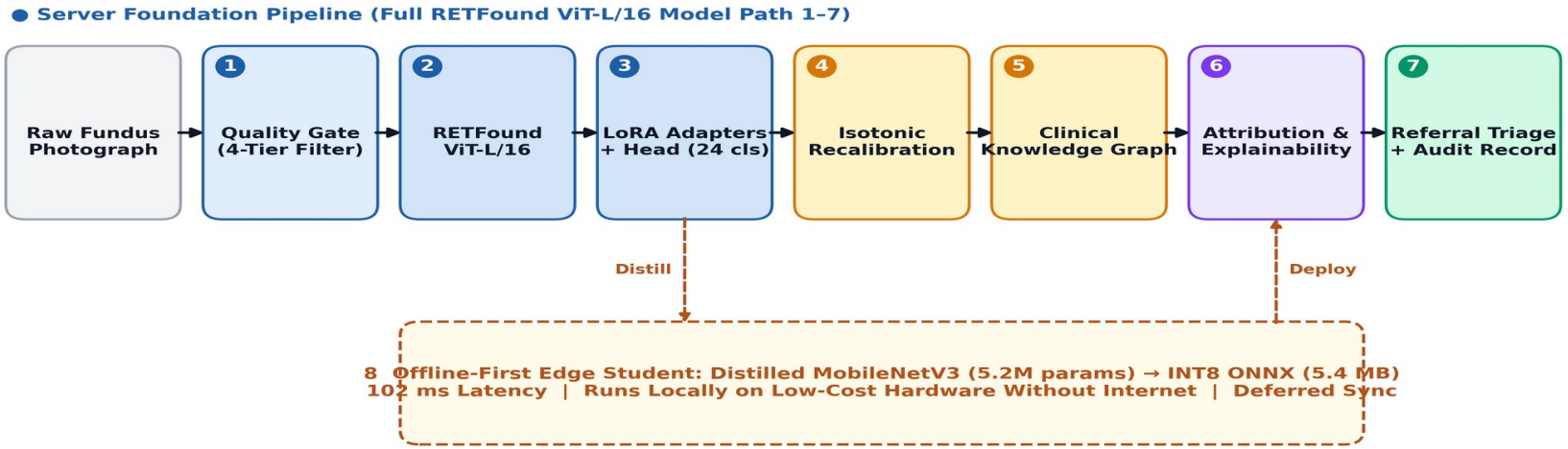
The Crisis: Avoidable Blindness in Uganda

2.2 Billion
Global Vision Impairment (WHO)
Over 50% is entirely preventable if diagnosed early.

< 45 Specialists
Ophthalmologists in Uganda
Serving 48M citizens — heavily concentrated in Mulago.

- **The Asymmetric Clinical Cost:** Diabetic retinopathy, glaucoma, and macular degeneration inflict irreversible blindness. Missing an insidious lesion strips individuals of agency, dignity, and livelihood.
- **The Frontline Bottleneck:** Primary nurses at Level III/IV health centres are frontline gatekeepers with zero ophthalmic diagnostic tools and minimal specialist training.
- **The Technological Dilemma:** Cloud-tethered 1 GB transformers fail when rural power drops and networks vanish. Screening requires an offline, highly specific rule-out sieve.

OptiscanAI: System Architecture



- **(1) Quality Gate:** Cascaded 4-tier intake filter (0.6s_stat + 0.4s_learned ≥ 0.70) rejects ungradable blurry photos.
- **(2–3) RETFound ViT-L + LoRA:** Pretrained on 1.6M retinal images; LoRA (r=16) adapts only 0.8% of weights (2.44M params).
- **(4–5) Calibration & Knowledge Graph:** Isotonic recalibration + 54-edge clinical ontology encoding AAO & Uganda MoH guidelines.
- **(6–7) Referral Triage & Audit:** Referral prioritization (Urgent vs Routine) with immutable cryptographic audit logging.

Offline Edge Distillation for Rural Clinics

5.4 MB
Model Size (INT8 ONNX)
Compressed 74.7% from 20.8 MB FP32 student.

102 ms
Median CPU Latency
Real-time inference on low-cost edge hardware.

99.6 %
Decision Concordance
Matches server FP32 decisions at $\tau = 0.50$.

- **The Reality of the Rural Clinic:** In Soroti, Gulu, or Karamoja, the electrical grid drops for days. Bandwidth to stream 15 MB uncompressed TIFF scans to cloud servers is pure fantasy.
- **Multi-Objective Knowledge Distillation:** A compact MobileNetV3-Large student (5.2M parameters) learns from the 305.7M-parameter RETFound ViT-L teacher via combined soft logits and feature alignment.
- **Zero-Network Autonomous Operation:** Triage executes 100% on-device. Encrypted audit records stage locally and sync opportunistically only when connectivity returns.

Rigorous Empirical Setup on RFMiD

640
Held-Out Test
Evaluated strictly post-convergence.

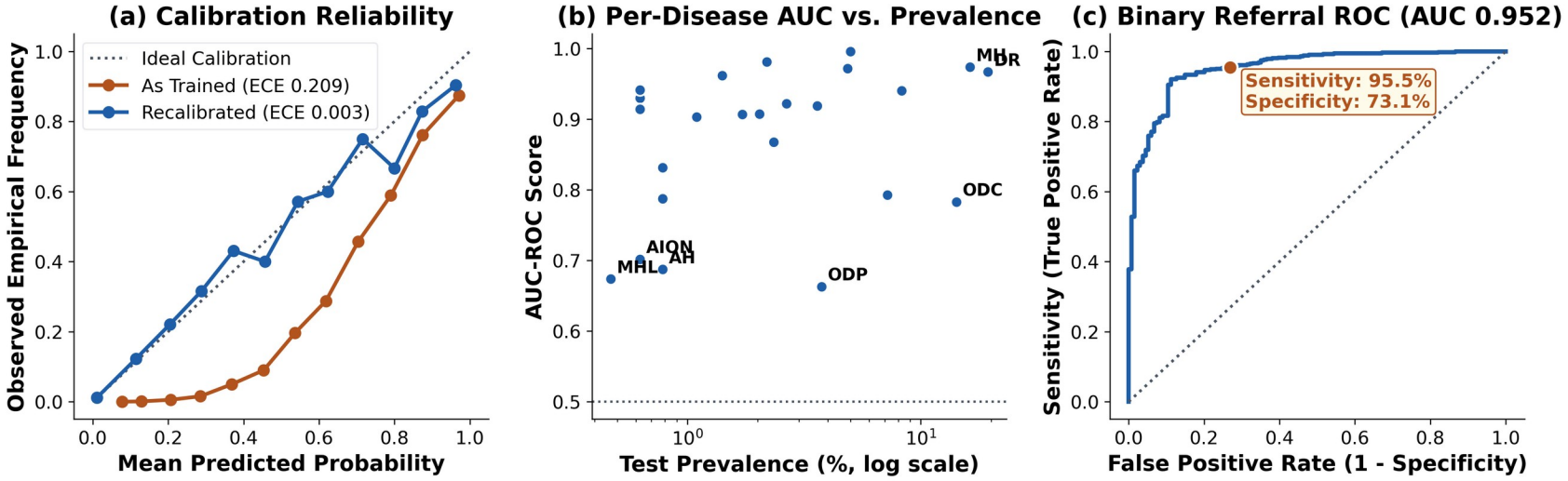
24
Retained Classes
Excluded rare labels (n < 10).

1,000
Bootstrap CIs
95% percentile confidence bounds.

CLAIM
Reporting Standard
Conforming to TRIPOD+AI 2024.

- **Strict Train / Val / Test Partition:** 1,920 train, 640 validation, and 640 evaluation images. All threshold selections and calibrations were locked strictly on validation data.
- **Input Normalisation Integrity:** Retained ImageNet normalisation because the foundation backbone was pretrained in that feature space; switching costs 0.054 AUC.
- **No Post-Hoc Cherry-Picking:** We report exact per-disease sensitivity, specificity, PPV, and false-negative rates with 95% bootstrap intervals across all 24 categories.

Multi-Disease Performance on RFMiD Test



- **Macro Discrimination (0.872 AUC):** Macro AUC reaches 0.872 (95% CI: 0.846–0.896) and Macro AUPRC reaches 0.357 across 24 multi-disease retinal categories.
- **High-Prevalence Excellence:** Myopia (0.996 AUC), Media Haze (0.974), Age-Related Macular Degeneration (0.972), and Diabetic Retinopathy (0.967 AUC, 0.873 AUPRC).
- **The Statistical Boundary (Petersonian Honesty):** Rare conditions with <20 test positives collapse: Optic Disc Pallor (0.663 AUC), Macular Hole (0.674). We refuse to claim clinical readiness where statistical power is absent.

The Silent Channel Defect & Remedy

0.209
Raw ECE (Failure)
Mean prob 0.251 vs 0.042 base rate. Extreme over-confidence.

5 Channels
Silenced Entirely
Static thresholding set fallback $\tau=0.95$: zero sensitivity.

0.003
Recalibrated ECE
Platt scaling restores reliable probabilities, 0 silent channels.

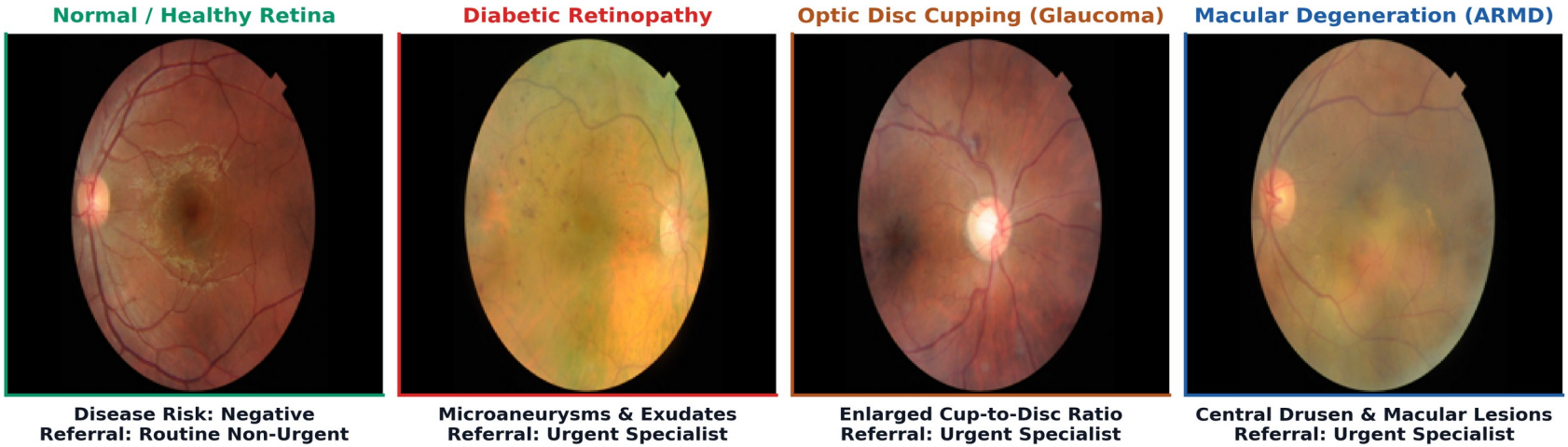
- **The Hidden Bug Nobody Measures:** Asymmetric loss clipping (0.05) and label smoothing prevent the network from ever outputting confident zeros. Minimum probability over the entire test split was 0.064!
- **The Fatal Interaction:** Searching for precision ≥ 0.10 on rare classes forced thresholds to the 0.95 ceiling, permanently muting 5 diagnostic channels (AH, ODP, PT, EDN, MHL).
- **The Monotone Solution:** Per-class Platt recalibration resolves ECE to 0.003 while preserving AUC ranking. Adopting a sensitivity target ≥ 0.90 eliminates all silent channels and reduces misses by 59%.

Ablation: Foundation vs. ImageNet

Model Architecture & Adaptation	Params (M)	Train. %	AUC-ROC	AUPRC
ResNet-50 (ImageNet Full FT)	23.56	100.0 %	0.867	0.336
EfficientNet-B3 (ImageNet FT)	10.73	100.0 %	0.819	0.377
ViT-B/16 (ImageNet Full FT)	85.82	100.0 %	0.871	0.472
RETFound ViT-L (Head Only)	2.44	0.8 %	0.877	0.360
RETFound ViT-L + LoRA (r=16)	**4.01**	**1.3 %**	**0.882**	**0.392**
OptiscanAI Deployed Checkpoint	**2.44**	**0.8 %**	**0.872**	**0.357**

- **Pretraining Efficiency:** Freezing RETFound and tuning only 0.8% params achieves 0.877 AUC, beating fully trained ImageNet CNNs. LoRA adds +0.005 AUC and +0.032 AUPRC.
- **The Fine-Tuning Nuance:** ViT-B/16 full fine-tuning attains higher AUPRC (0.472 vs 0.392). Retinal foundation models give parameter efficiency, but full fine-tuning retrieves rare positives better.
- **Normalisation Sensitivity:** Substituting RFMiD channel statistics for ImageNet normalisation costs 0.054 AUC and 0.107 AUPRC. Preserving pretraining distribution alignment is critical.

Clinical Screening Gallery: Real Cases



- **Healthy Control Eye:** Disease Risk Negative → Routine follow-up. Prevents needless tertiary hospital visits.
- **Diabetic Retinopathy (DR):** Microaneurysms detected → Automated Urgent Referral for laser photocoagulation.
- **Optic Disc Cupping (ODC):** C/D ratio > 0.6 detected → Urgent triage for intraocular pressure and visual field testing.
- **Macular Degeneration (ARMD):** Central drusen lesions detected → Urgent ophthalmology referral for anti-VEGF intervention.

Prevalence Shift: Benchmark vs. Clinic

Screening Caseload / Setting	Prevalence	PPV	NPV	Referral %
RFMiD Benchmark (Held-Out Test)	**79.0 %**	**0.931**	**0.810**	**81.0 %**
District Hospital Caseload	20.0 %	0.470	0.985	41.0 %
Rural Primary Care Clinic	10.0 %	0.283	0.993	34.0 %
Community Mobile Screening	5.0 %	0.158	0.997	30.0 %

- **The Referral Bottleneck Tragedy:** At 5% community prevalence, PPV drops from 93.1% to 15.8%. Sizing referral clinics on benchmark numbers will overburden ophthalmologists by 3x–6x with non-pathology cases.
- **The True Role: A Safe Rule-Out Sieve:** Negative Predictive Value (NPV) exceeds 99.7% in low prevalence. OptiscanAI is not an autonomous doctor; it is a high-confidence sieve to protect specialists.

Limitations and Conclusion

LIMITATIONS & PRECONDITIONS

- **Zero Ugandan Patient Images:** Evaluated on international benchmark (RFMiD). Multi-centre clinical trials in Ugandan district hospitals remain mandatory.
- **Handheld Optical Shift:** Validation on table-top fundus cameras cannot guarantee performance on low-cost smartphone adapters.
- **Regulatory Preconditions:** Requires Uganda National Drug Authority (NDA) and Data Protection Act (2019) clearance before clinical trials.

KEY SCIENTIFIC TAKEAWAYS

0.872 AUC / 0.952 Triage
Robust Ranking & Referral
Evaluated strictly on 640 held-out test images.

0.003 Calibrated ECE
Silent Channels Resolved
Platt recalibration restores 5 disabled disease channels.

5.4 MB / 102 ms
Offline Edge Distillation
Runs autonomously on mobile hardware in rural health centres.

▸ **OptiscanAI establishes a disciplined scientific framework that separates benchmark excellence from clinical deployment reality.**

THANK YOU

Questions & Discussion

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Live Interactive Prototype: https://landwind22-retinal-screening.hf.space